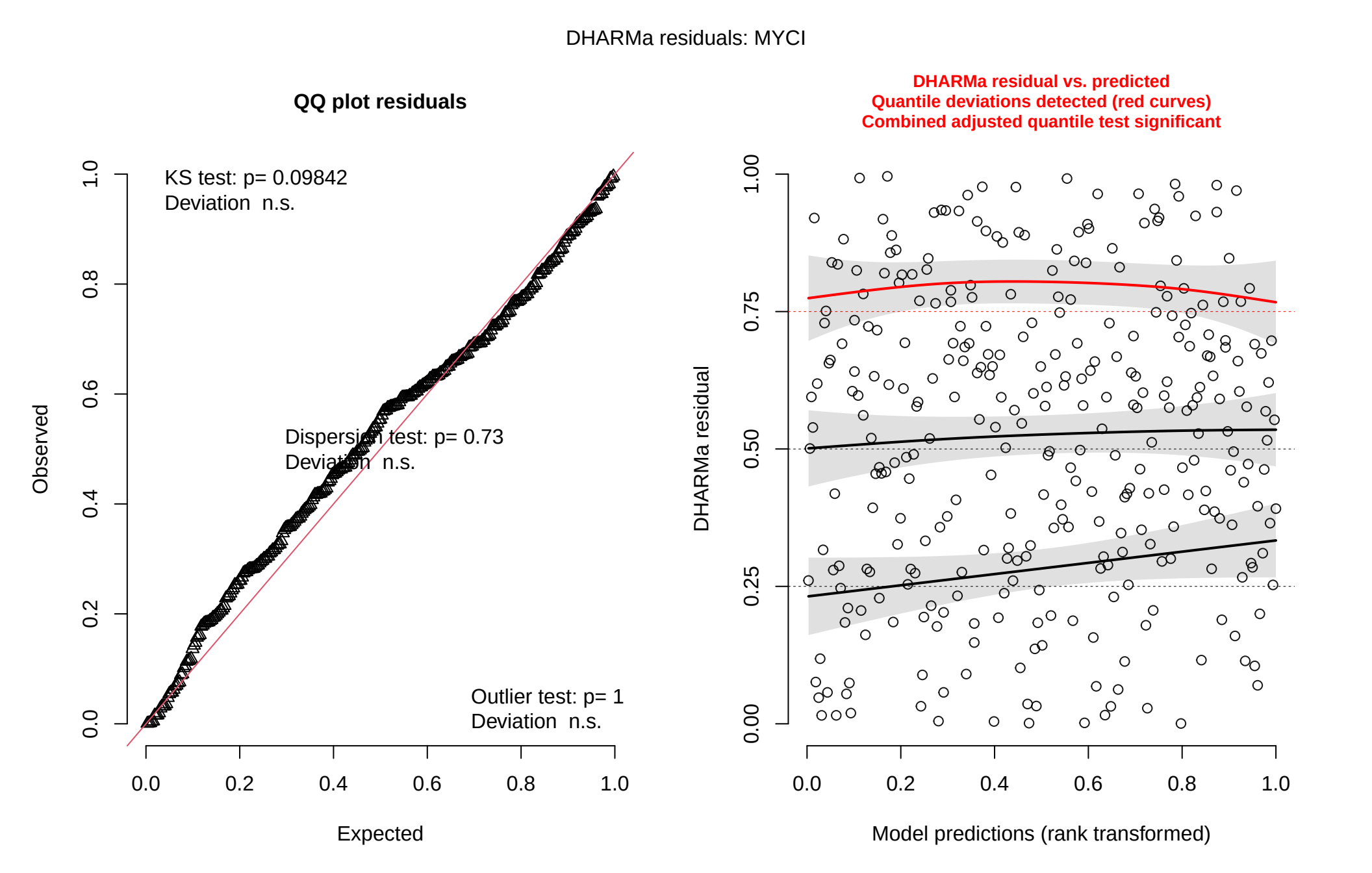
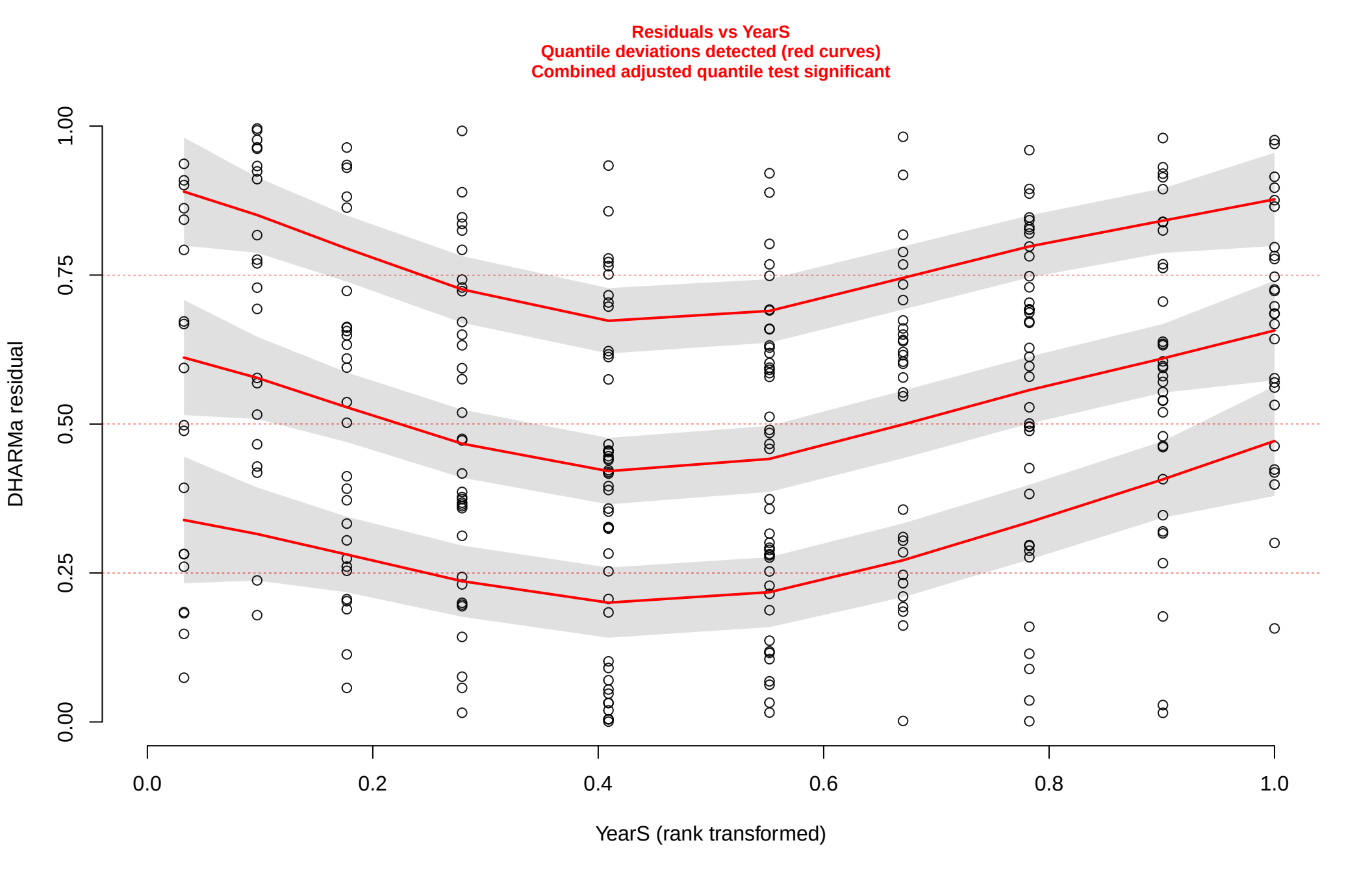
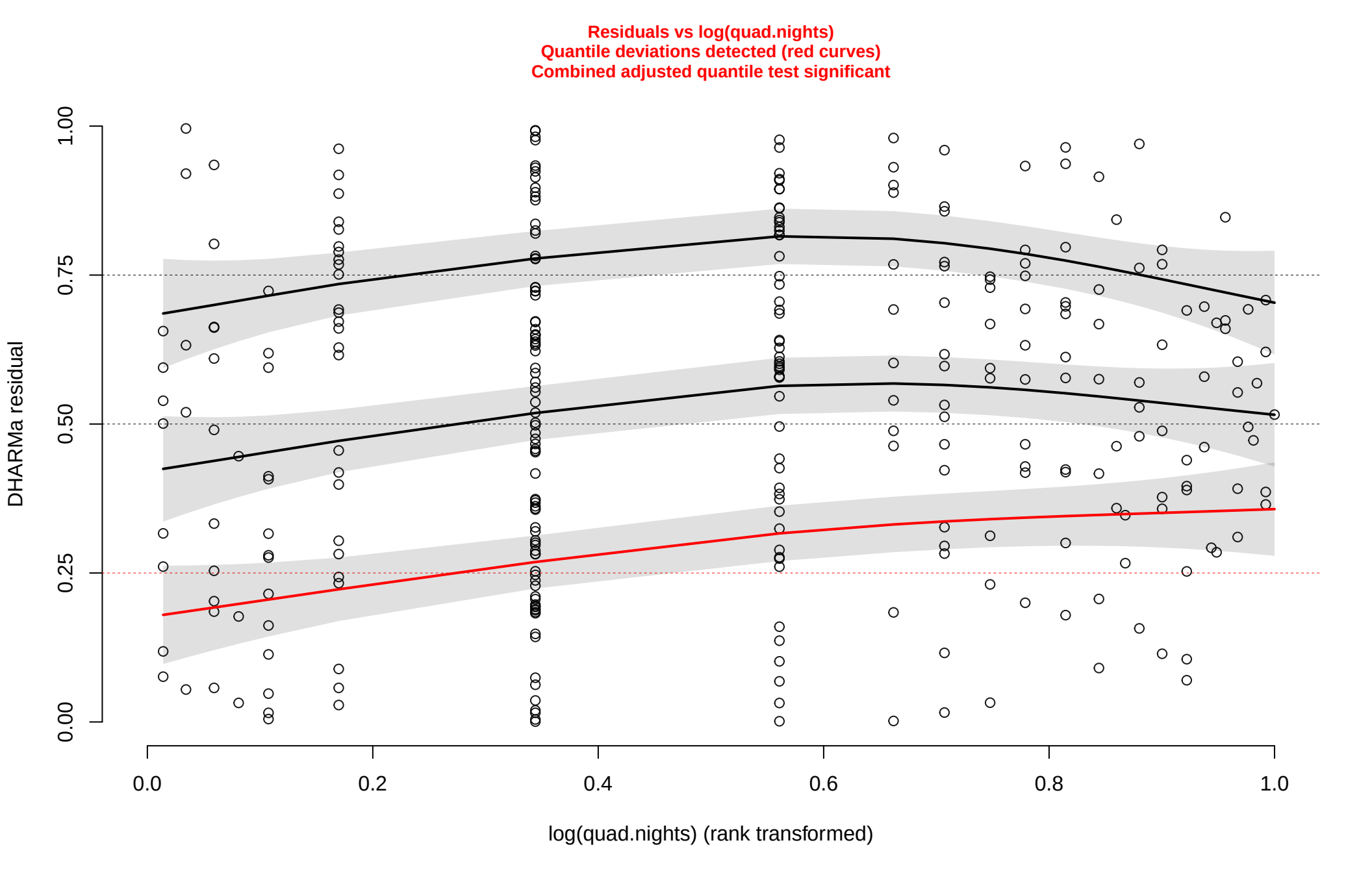
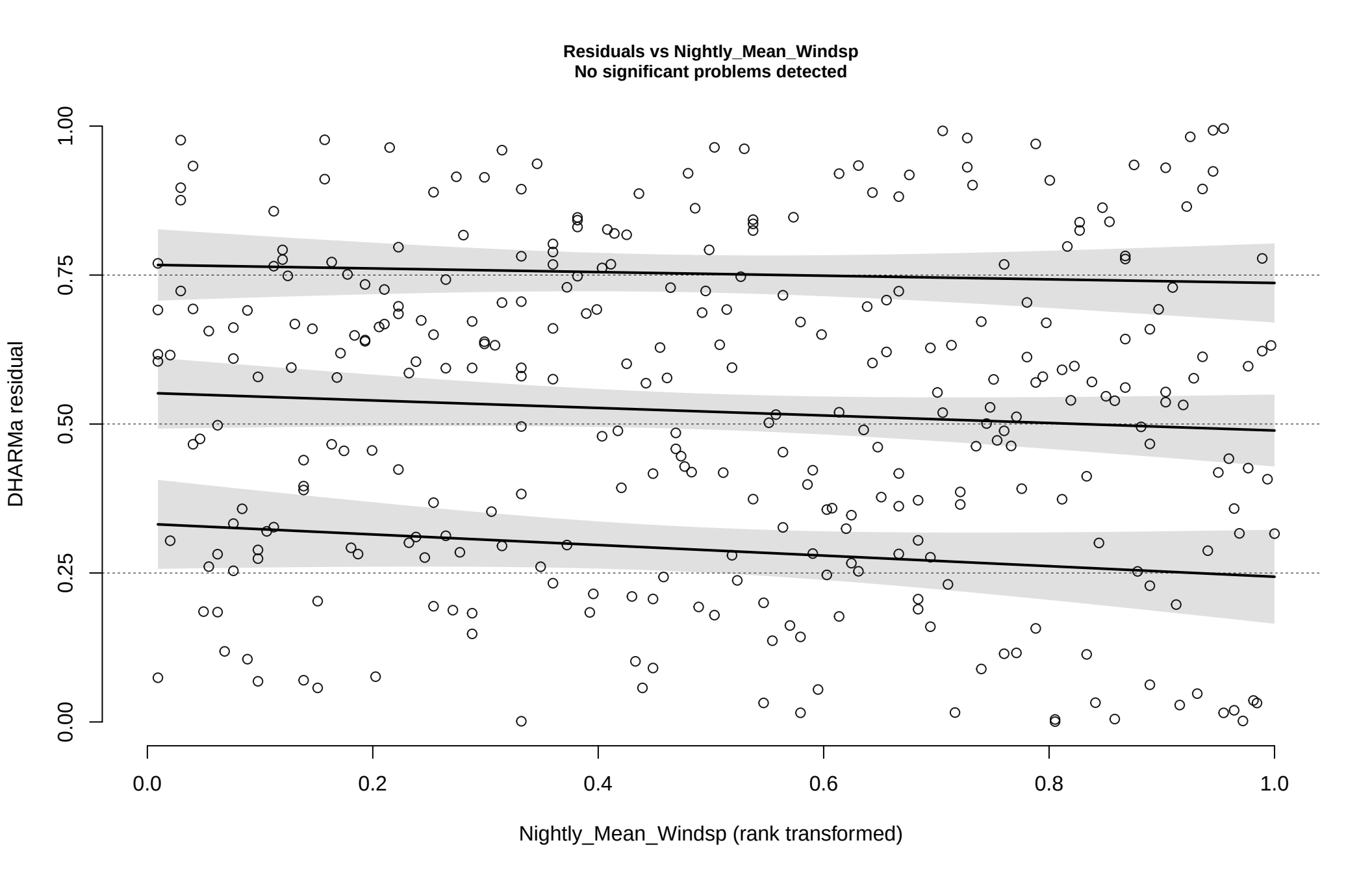
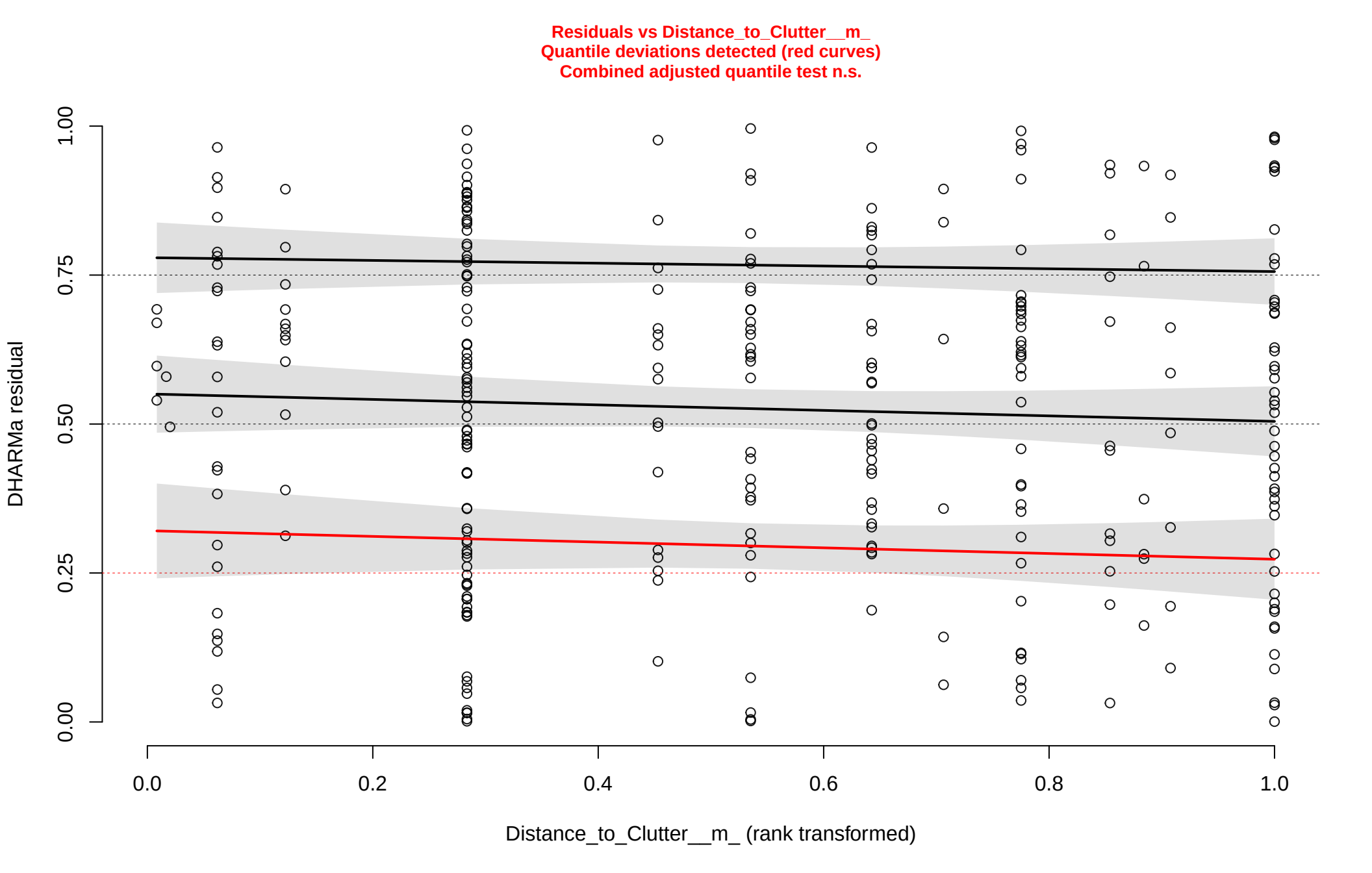
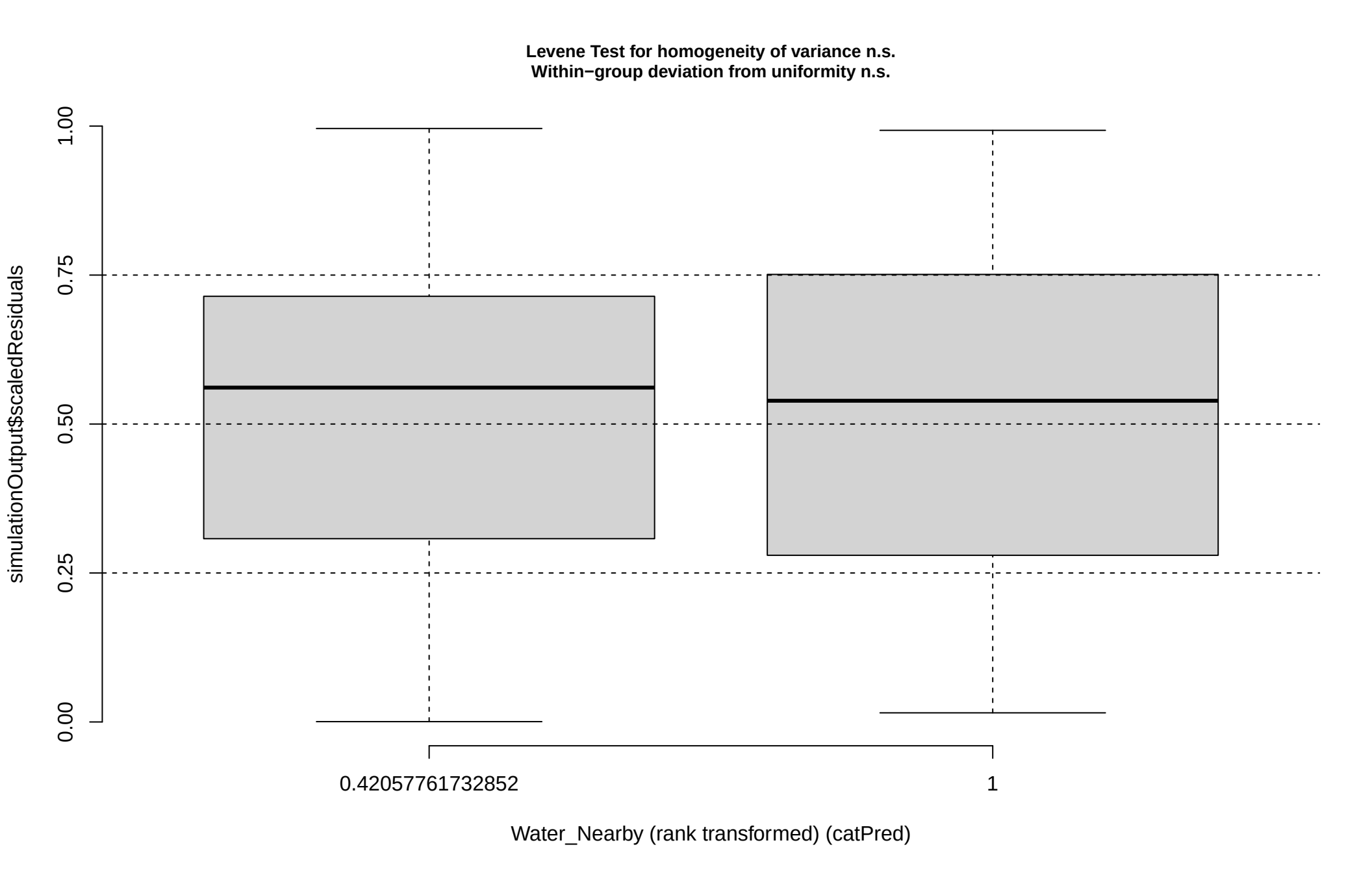

DHARMa residuals: MYCI
QQ plot residuals
KS test: p= 0.09842
Deviation n.s.
Dispersion test: p= 0.73
Deviation n.s.
Outlier test: p= 1
Deviation n.s.
Observed
Expected
0.0
0.2
0.4
0.6
0.8
1.0
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)
0.00
0.25
0.50
0.75
1.00

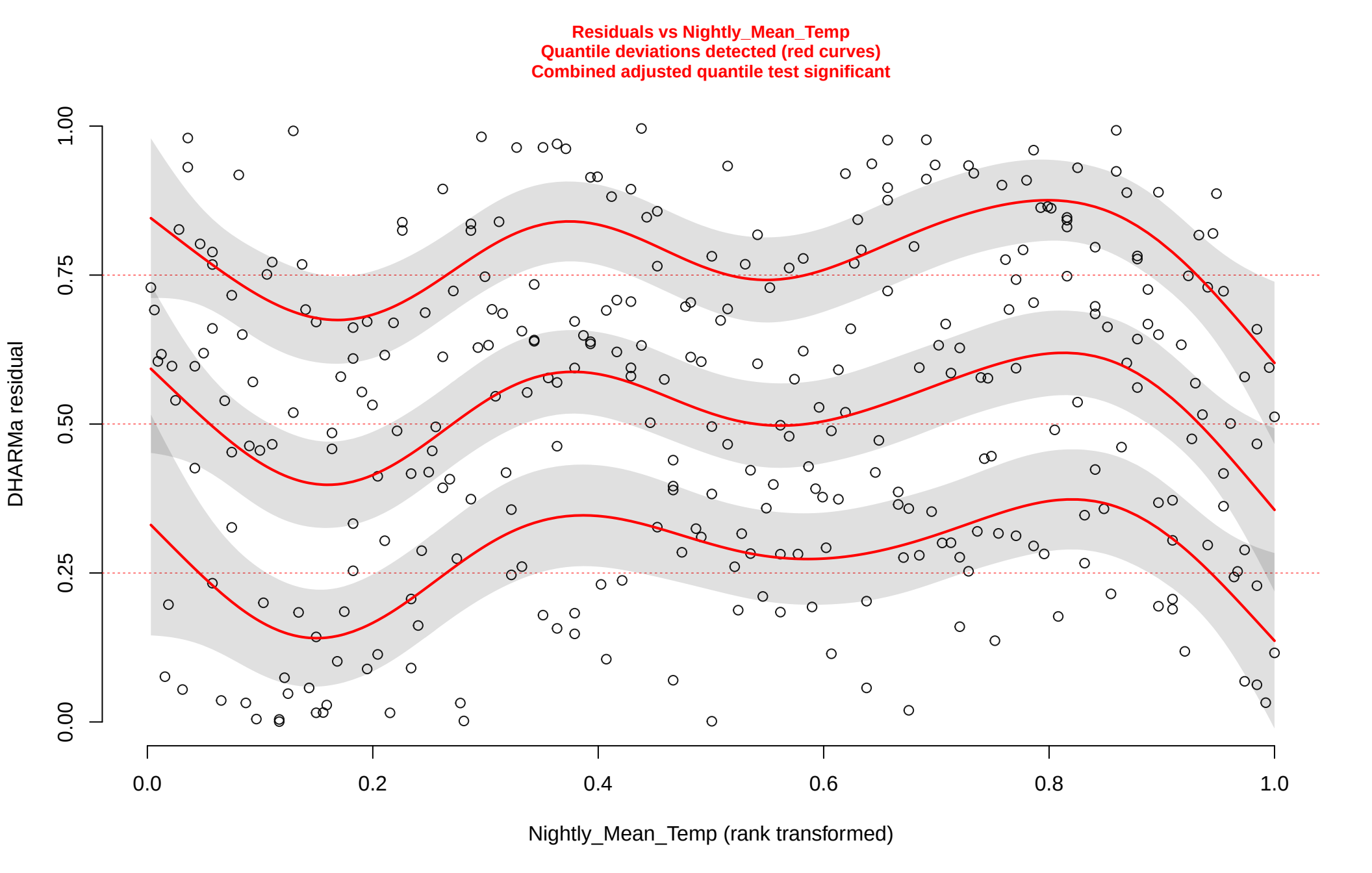

Residuals vs YearS
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

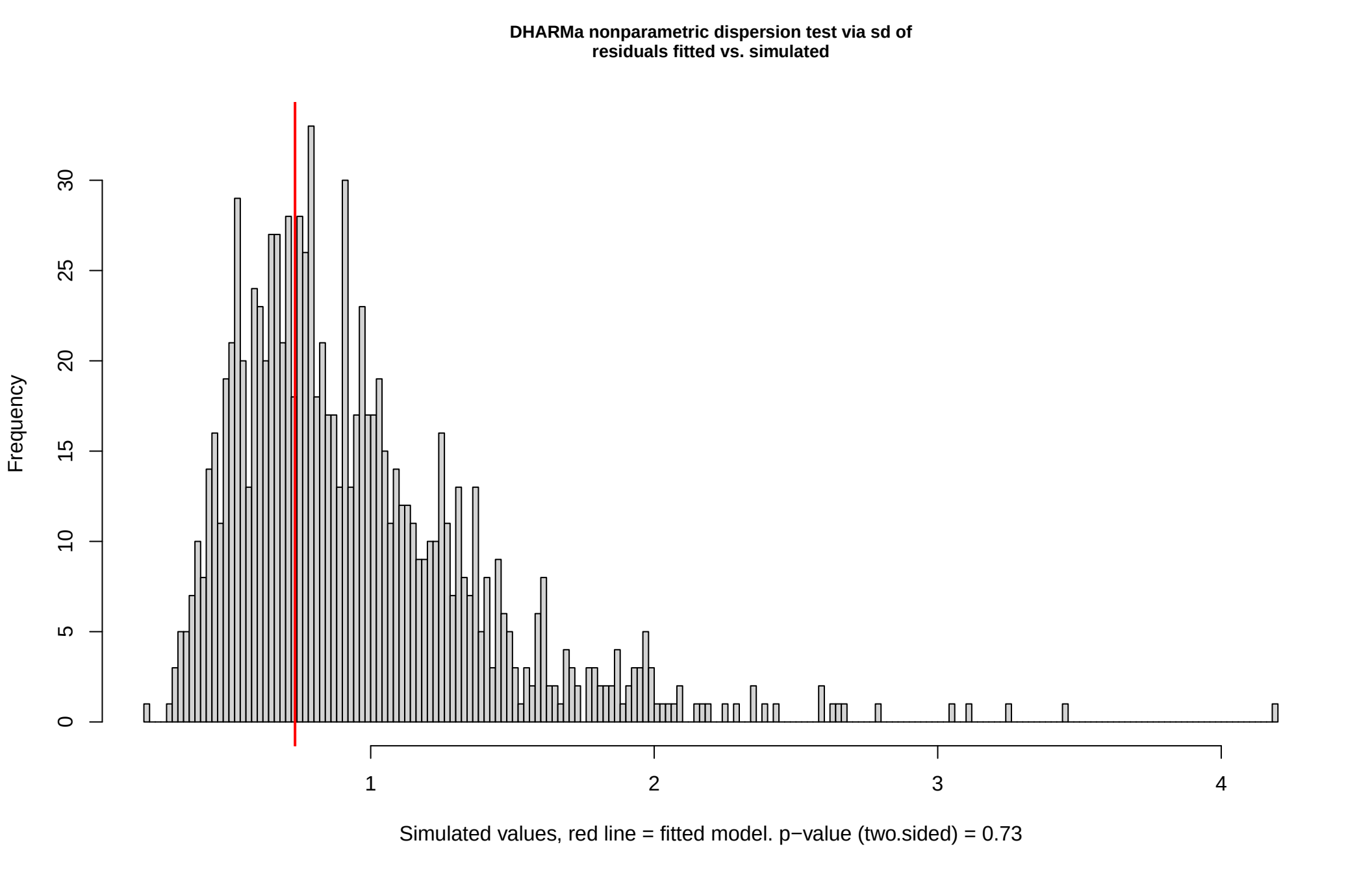

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

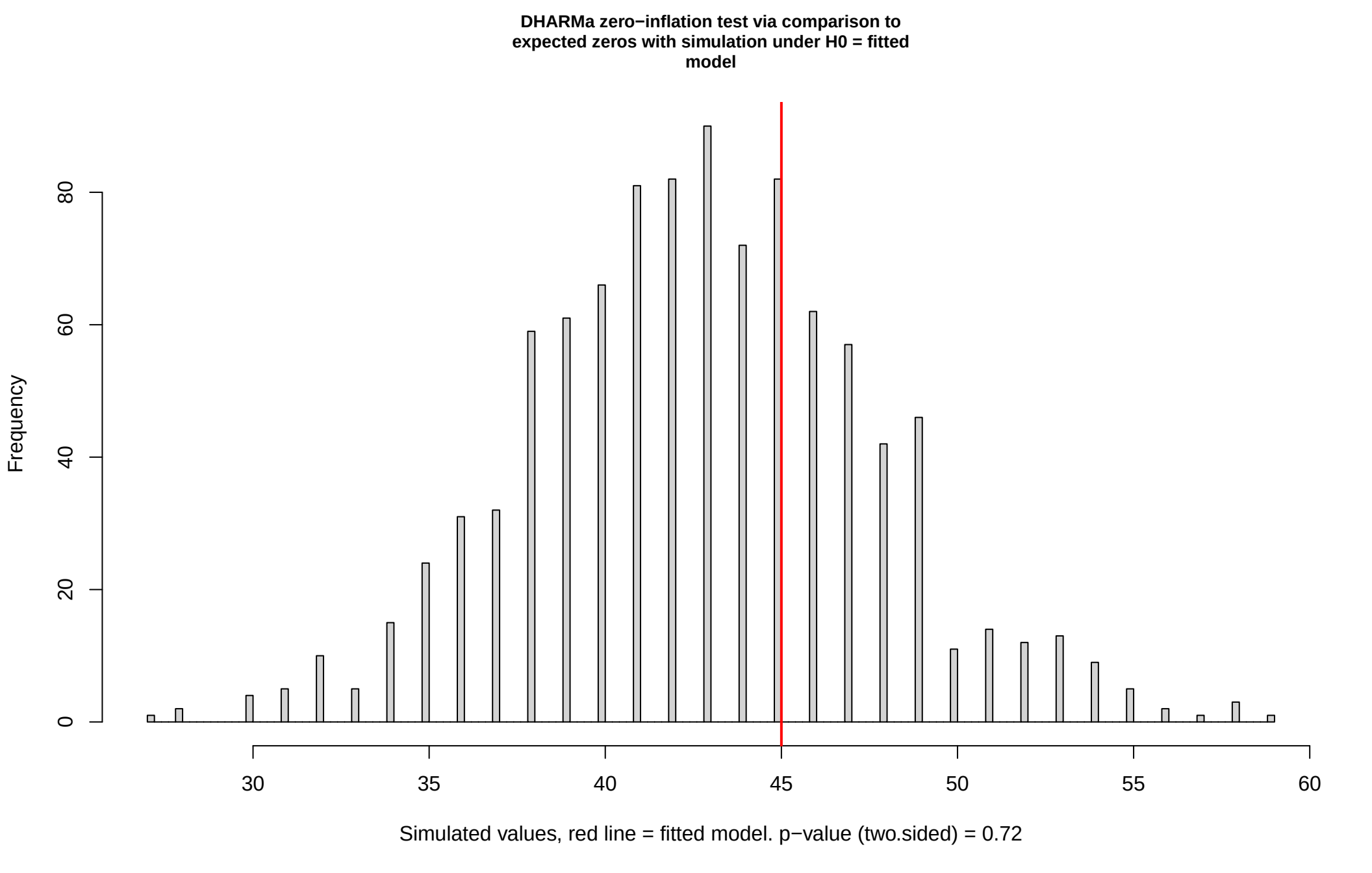

Residuals vs Nightly_Mean_Windsp
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)

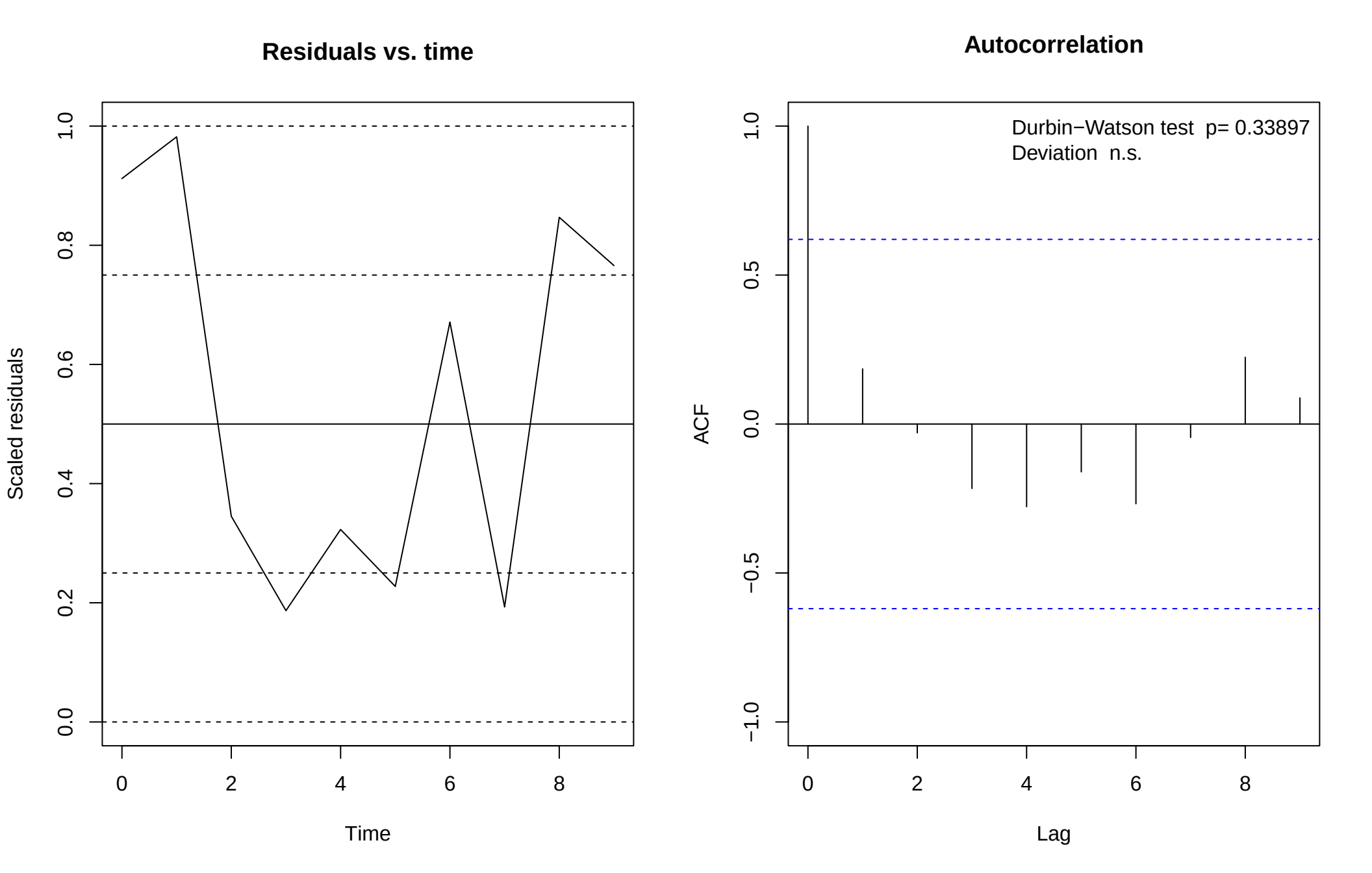

Residuals vs Distance_to_Clutter__m_
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

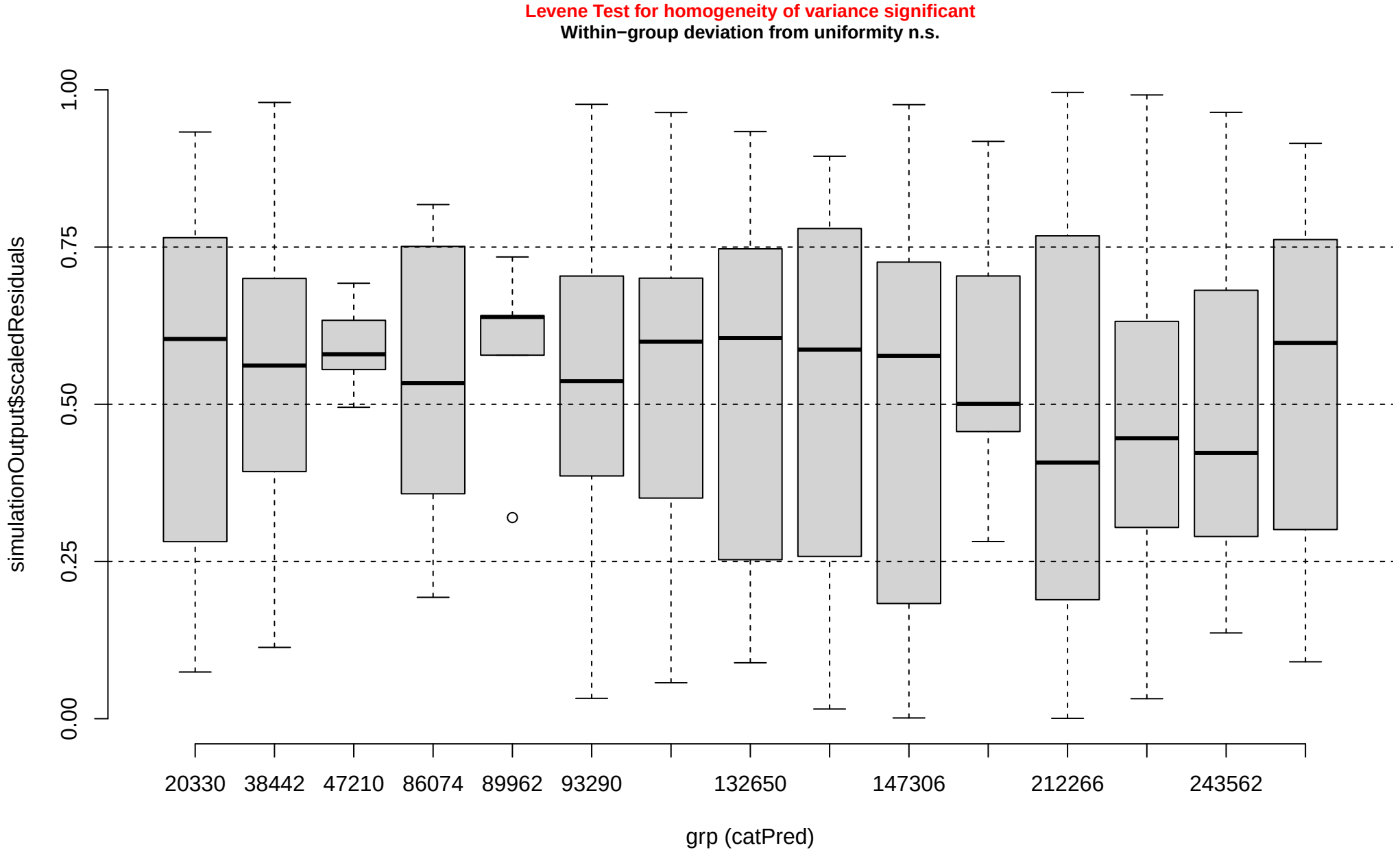

Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
0.42057761732852
1
Water_Nearby (rank transformed) (catPred)

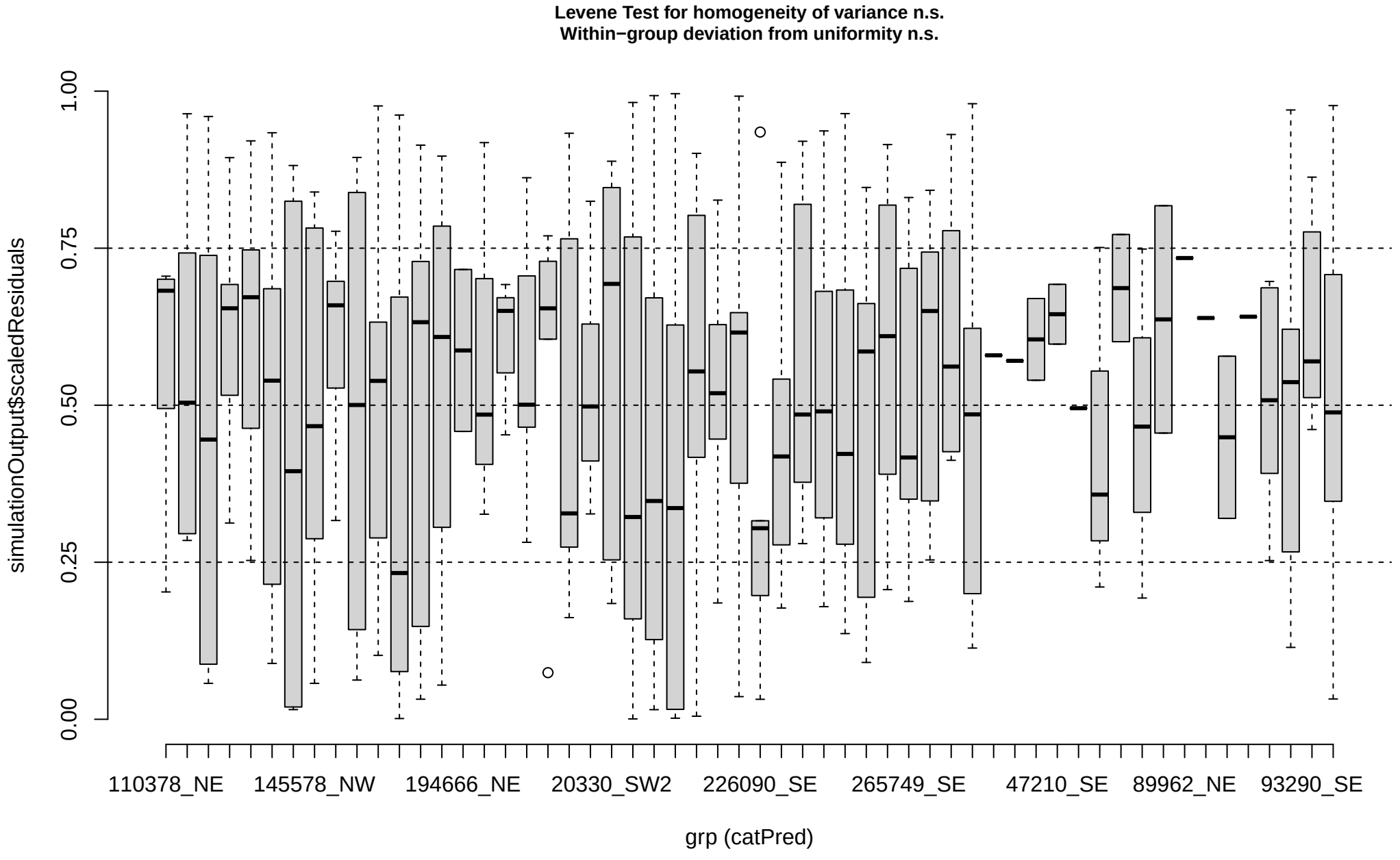

Residuals vs Nightly_Mean_Temp
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

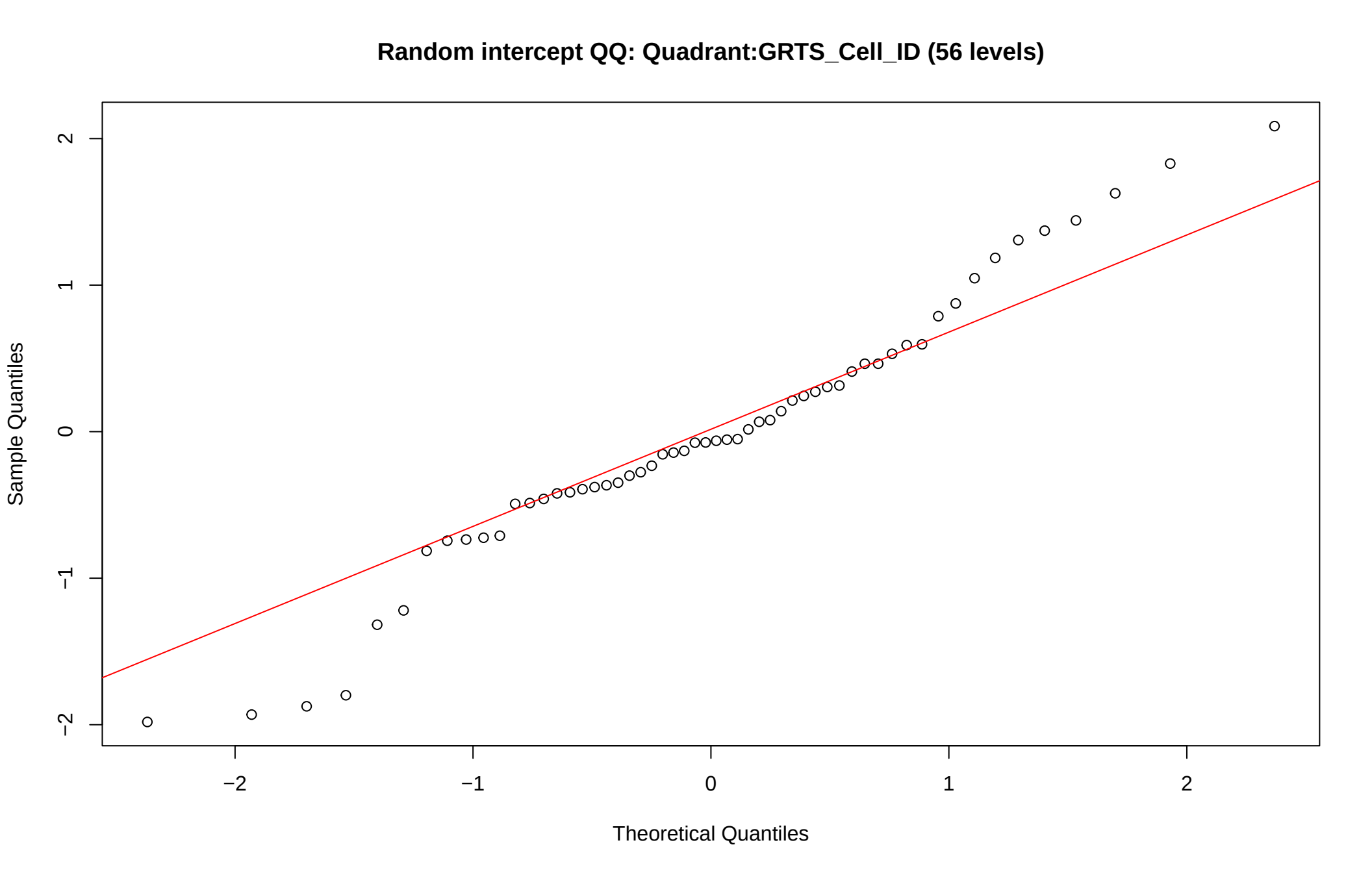

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
20
25
30
1
2
3
4
Simulated values, red line = fitted model. p−value (two.sided) = 0.73

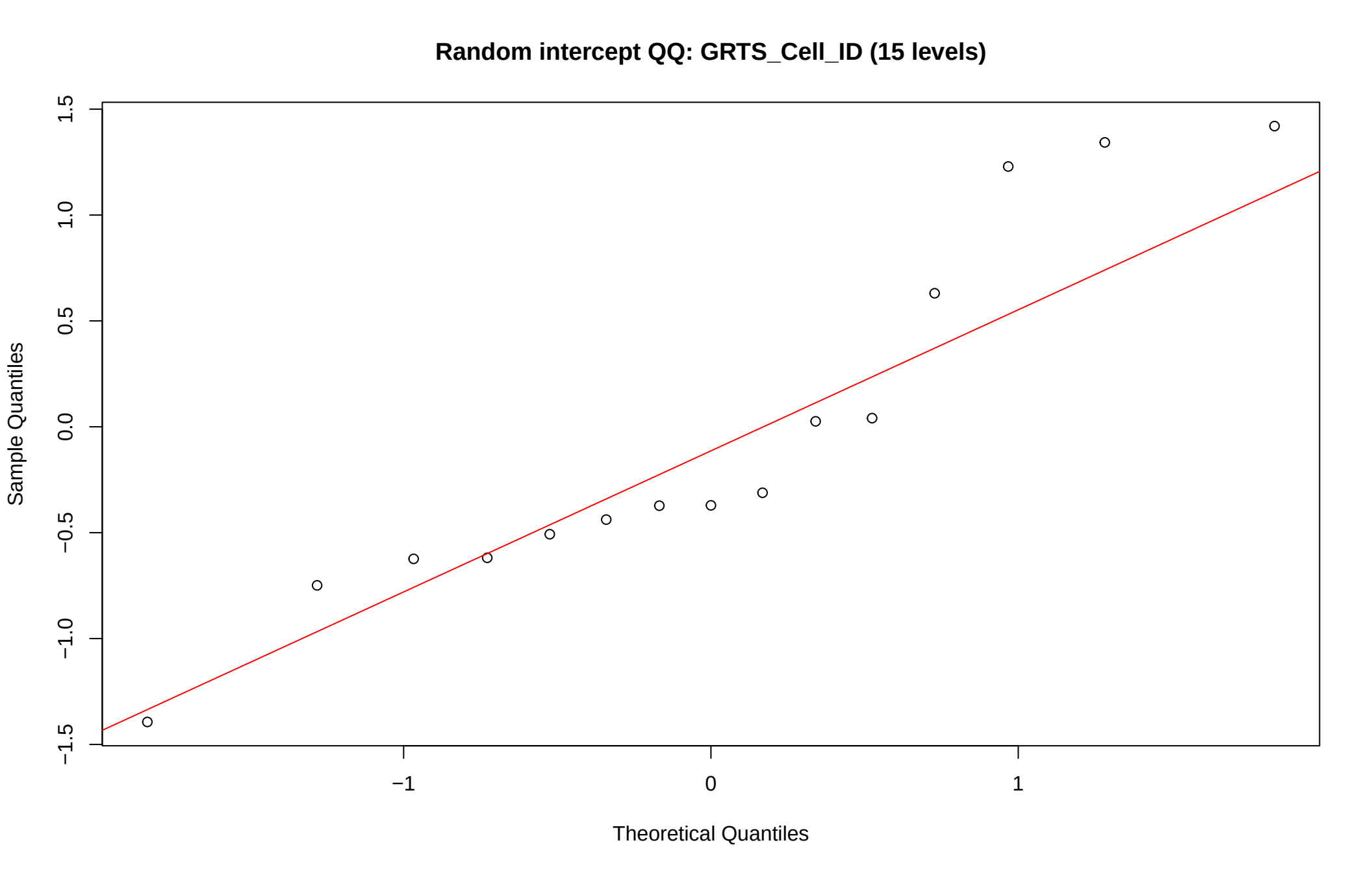

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
20
40
60
80
30
35
40
45
50
55
60
Simulated values, red line = fitted model. p−value (two.sided) = 0.72


Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time
Autocorrelation
ACF
Durbin−Watson test p= 0.33897
Deviation n.s.
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance significant
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
20330
38442
47210
86074
89962
93290
132650
147306
212266
243562
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
110378_NE
145578_NW
194666_NE
20330_SW2
226090_SE
265749_SE
47210_SE
89962_NE
93290_SE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (56 levels)
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (15 levels)
Sample Quantiles
Theoretical Quantiles
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1
0
1